RESEARCH PRESENTATION

Smart UV Sterilization Safety System

Using Multi-Layer Protection Mechanisms

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UV-C Sterilization • PIR Human Detection • Door Interlocking • Arduino Uno • Emergency Shutdown

Introduction

- UV-C radiation (200–280 nm) damages microbial DNA/RNA, offering a fast, chemical-free, residue-free alternative to manual cleaning and chemical disinfectants.
- Adoption has accelerated across hospitals, laboratories, airports, public transport, and food processing — driven further by COVID-19 disinfection demand.
- Direct UV-C exposure causes severe skin and eye damage, making user safety as critical as disinfection performance.
- Most existing systems rely on a single safety layer (e.g., a timer or one occupancy sensor) — a fragile design if that one mechanism fails.

UV-C Wavelength Range

200–280 nm

Why safety matters

- Damages DNA/RNA of bacteria, viruses, fungi
- Leaves no chemical residue
- Causes serious skin & eye injury on direct exposure
- Demands engineered, multi-layer safeguards

Problem Statement

Existing UV-C sterilization systems prioritize disinfection efficiency over comprehensive user protection — leaving real exposure risks unaddressed.

Single Point of Failure

Many systems depend on manual switching, timers, or one occupancy sensor — if that single mechanism fails or responds late, exposure occurs.

No Physical Interlocking

Door status is rarely verified or enforced; accidental opening during a cycle can directly expose a user to UV-C radiation.

No Automatic Locking

Chamber doors are almost never auto-locked during operation, so the chamber can be opened mid-cycle with no mechanical safeguard.

Limited Warning & Monitoring

Pre-activation alerts and continuous real-time monitoring throughout the cycle are only partially implemented, if at all.

Core problem: no unified, low-cost embedded system combines door verification, human detection, auto-locking, warnings, and emergency shutdown.

Literature Review

KEY FOCUS AREA

Refs [1]–[5]

UV-C Sterilization Systems

Chambers, portable devices & automated UV-C systems prove effective, chemical-free disinfection in hospitals, labs, transport & healthcare.

Refs [1]–[5]

Refs [6]–[10]

Robotic & Autonomous Disinfection

Robots use autonomous navigation, obstacle avoidance, occupancy detection & path planning to disinfect hospitals, radiology depts, airports.

Refs [6]–[10]

Refs [11]–[15]

Human Safety & Exposure Prevention

Occupancy detection, motion sensing, wireless UV dose monitoring, UV sensors & Far UV-C studied to reduce exposure.

Refs [11]–[15]

Research Gap Analysis (Table I)

Feature	Existing Systems	Proposed System
Human Presence Detection	Partial	✓
Automatic Door Locking	✗	✓
Multi-Layer Safety Framework	✗	✓
Emergency Shutdown	Partial	✓

The Gap We Fill

No prior work unifies door interlocking, auto-locking, human detection, warnings, real-time monitoring AND emergency shutdown in one low-cost embedded platform — each study isolates 1–2 mechanisms only.

Literature Review — Comparative Landscape

Table III — Proposed system vs. representative classes of UV systems reported in literature

Feature	Existing UV Systems	Safety Robots	Monitoring Systems	Proposed System
UV-C Sterilization	✓	✓	✓	✓
Human Detection	Partial	✓	✓	✓
Door Interlocking	✗	✗	Partial	✓
Automatic Door Locking	✗	✗	✗	✓
Warning Alerts	Partial	Partial	Partial	✓
Real-Time Monitoring	Partial	✓	✓	✓
Emergency Shutdown	Partial	Partial	Partial	✓
Low-Cost Embedded Design	Partial	✗	Partial	✓

Proposed system is the only one in this comparison checking every column — the basis for the multi-layer framework introduced next.

Objectives

Multi-layer safety framework

Design a system combining human detection, door interlocking, auto-locking, warnings, monitoring & emergency shutdown.

Real-time condition verification

Continuously verify door closure (reed switch) and human presence (PIR) before and during sterilization.

Automatic emergency protection

Immediately deactivate the UV-C lamp and terminate the cycle whenever an unsafe condition is detected.

Low-cost embedded platform

Build an affordable, Arduino Uno–based architecture suitable for small-scale deployment.

Broad applicability

Deliver a solution viable for hospitals, laboratories, educational institutions, offices, and households.

Validate through testing

Empirically confirm functional reliability and safety performance across defined test scenarios.

Proposed Methodology — System Architecture

Centralized embedded design — Arduino Uno as main controller, organized into three functional layers

INPUT LAYER

PIR Sensor

Human presence detection

Magnetic Reed Switch

Door open/closed verification

PROCESSING LAYER

Arduino Uno

Decision-making unit

Safety Logic

Verifies door + human conditions before/throughout cycle

OUTPUT LAYER

Relay + UV-C Lamp

Powers sterilization

Servo Motor

Auto lock / unlock door

LEDs + Buzzer

Status & warning alerts

Proposed Methodology — 5-Stage Safety Framework

1

Door closure verification

Magnetic reed switch confirms the chamber door is fully closed.

2

Human presence detection

PIR sensor checks for nearby occupants before proceeding.

3

Pre-sterilization warning

Buzzer + LED indicators alert users prior to UV-C activation.

4

Automatic door locking

Servo motor locks the chamber to prevent mid-cycle opening.

5

Continuous monitoring + emergency shutdown

Sensors polled throughout; any unsafe state instantly kills the lamp, unlocks the door, and sounds the alarm.

Operational flow: Place object → close door → verify door & absence of humans → warn → lock door → activate UV-C lamp via relay → monitor continuously → auto shutdown on door-open/human-detected → unlock & signal completion when timer elapses safely.

Results and Discussion

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Functional test scenarios passed (Table II)

Test Scenario	Result
Door open → lamp stays OFF	Pass
Human detected → lamp stays OFF	Pass
Pre-sterilization warning (buzzer + LED)	Pass
Automatic door locking (servo)	Pass
Emergency shutdown on unsafe condition	Pass
Sterilization complete → unlock + indicate	Pass

Key Findings

- UV-C lamp never activated while the door was open or a human was detected.
- Unexpected door-opening or human detection mid-cycle triggered immediate relay shutdown + alarm.
- Servo motor reliably locked/unlocked the door every cycle; LEDs correctly tracked all states (ready, in-progress, emergency, complete).
- No significant communication delays or component failures during repeated trials.

Compared to literature

- Outperforms existing systems on door interlocking & automatic locking — features marked × or Partial in Table III.
- Matches or exceeds robotic/monitoring systems on real-time monitoring while remaining a low-cost embedded design.
- Only configuration tested that satisfies every safety column simultaneously.

Conclusion

A working, low-cost, Arduino-based safety platform that proves multi-layer protection is achievable without sacrificing sterilization performance.

- PIR + reed switch + servo + relay + Arduino Uno combine into one coordinated, autonomous safety platform.
- Sterilization activates only under verified safe conditions and halts instantly when conditions change.
- Experimental evaluation confirmed reliable operation across all 10 test scenarios with no failures.
- Practical and economical for hospitals, laboratories, education, offices, and households alike.
- Lays the foundation for future IoT-enabled, dose-monitored, AI-assisted sterilization systems.

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Full list of 30 references available in the source paper.